

Project Design Phase

Problem – Solution Fit Template

Date	29 June 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	2 Marks

Problem – Solution Fit:

The Problem–Solution Fit process helps identify whether the proposed solution effectively addresses the real problems faced by customers. For the Smart Lender project, the primary challenge is the lengthy and manual loan approval process, which often results in delays, human errors, and inconsistent credit evaluation. By understanding customer needs, pain points, and existing limitations, our team designed an AI-powered solution that automates loan eligibility prediction using machine learning algorithms. This approach enables financial institutions to make faster, more accurate, and data-driven lending decisions while improving customer experience and reducing credit risk.

Purpose:

The Smart Lender project aims to provide an intelligent loan approval system that accurately predicts the creditworthiness of applicants using machine learning techniques. The solution helps banks and financial institutions reduce manual effort, minimize loan default risks, improve processing speed, and enhance overall decision-making. The Problem–Solution Fit template validates that the proposed solution aligns with customer requirements and delivers measurable business value.

Market & Customers	1.CUSTOMER SEGMENTS CS • Banks and Financial Institutions • Credit Officers • Loan Applicants • Financial Analysts • NBFCs (Non-Banking Financial Companies)	 2.JOBS TO BE DONE / PROBLEM JP • Evaluate loan applications quickly. • Reduce manual loan approval time. • Minimize loan default risk. • Improve decision-making accuracy. • Process large numbers of loan applications efficiently.	 3.CUSTOMER PAINS CP • Time-consuming manual verification. • Human errors during evaluation. • High risk of approving defaulters. • Delayed loan processing. • Difficulty in handling large application volumes.	Market & Customers
	 4.EXISTING ALTERNATIVES EA • Manual credit assessment. • Rule-based loan approval systems. • Spreadsheet-based evaluation. • Traditional banking software.	 5.CUSTOMER CONSTRAINTS CC • Limited credit history of applicants. • Incomplete or incorrect applicant data. • Need for quick loan decisions. • Data security and privacy concerns.	 6.AVAILABLE SOLUTIONS AS Smart Lender – AI-Powered Loan Approval Prediction System • Predicts loan approval using Machine Learning. • Uses XGBoost, Random Forest, Decision Tree, and KNN models. • Provides real-time predictions. • Reduces manual effort. • Improves approval accuracy.	
 7.TRIGGERS TR • Applicant submits a loan application. • Bank receives a new loan request. • Credit officer needs faster evaluation. • High volume of loan applications.	 9.CHANNELS OF DISTRIBUTION CH • Flask Web Application • IBM Cloud Deployment • Bank Internal Portal • Desktop Web Browser • Mobile Web Browser	 COST STRUCTURE CT • Cloud hosting and infrastructure costs • Model training and maintenance • Data storage and security • Development and operational costs	Validation & Growth	
 8.VALUE PROPOSITION VP • Faster loan approval decisions. • Higher prediction accuracy. • Reduced non-performing assets (NPAs). • Automated credit risk assessment. • Improved customer experience.	 REVENUE STREAMS RS • Subscription / SaaS model for banks and NBFCs • Licensing fees • Customization and integration charges • Premium analytics and reports	Validation & Growth		

